Homework 3 - Querying the UCSC database and Analyzing the data

Graded

Student

Kalkidan Tadese

Total Points

10 / 10 pts

Autograder Score

0.0 / 0.0

Question 2

1 — **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 0.4 pts** Didn't read gene list from file
- **– 0.4 pts** Multiple transcripts per gene
- **– 0.3 pts** Didn't use gene list in SQL query
- **– 0.4 pts** Loaded table into RAM. Query should be done within SQL.

Question 3

2 — **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 0.4 pts** Reads should be within the transcription boundaries, no partial overlap
- **– 0.4 pts** Queried refGene table instead of refGeneCoor.
- **– 0.2 pts** Full table name instead of just the tissue
- **– 0.4 pts** Counting multiple positions per gene.
- **– 0.4 pts** Loaded tables to RAM. Query should be done within SQL. Use count(*)

Question 4

3 — **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 0.2 pts** quoted output
- **– 0.2 pts** Disconnect database
- **– 0.2 pts** Full table name instead of just tissue
- **– 0.4 pts** Applied mean/sd function to whole column, not per gene

Question 5

4

2 / 2 pts

✔ – 0 pts Correct

– 0.4 pts Gene name should be the x axis, tissue should be fill color

– 0.2 pts Full table name instead of just the tissue

– 0.2 pts Overlapping bars

Autograder Results

This assignment does not have an autograder configured.

Submitted Files

Kalkidan_Tadese (1).ipynb

Download

In [1]:

```
#install sqlite3 and pandas
import sqlite3
import pandas as pd
```

In [2]:

```
gene_names = pd.read_csv("Gene_names.txt", header=None)
#reading in the gene_names file, specifying no header
gene_names = gene_names[0].tolist() #turning the values generated in the
unnamed column into a list
gene_names #print list
```

Out [2]:

```
['PADI2',
 'HSPB7',
 'PLA2G2A',
 'AGMAT',
 'FABP3',
 'FCN3',
 'SERINC2',
 'MASP2',
 'STMN1',
 'NBL1',
 'LDLRAD2',
 'RBP7',
 'SLC2A5',
 'FBXO2',
 'MXRA8',
 'ANKRD65',
 'CDC42',
 'EPHB2',
 'PRKCZ-AS1',
 'HSPG2',
 'PHACTR4',
 'CASP9',
 'ATAD3B',
 'KIAA2013',
 'AURKAIP1']
```

In [3]:

```
con = sqlite3.connect("hg19.sqlite") #connect to sqlite database
c = con.cursor() # create pointer/cursor object to interact with db
```

In [4]:

```
placeholders = ", ".join(["?" for _ in gene_names])
#list comprehension - for every item in gene_names, store it as "?"
# and join the "?" into a list with a comma and a space after each one

#subselect to extract specific columns from refGene and partition (split into
smaller bins) the items based on their name2
#order the partitioned items and assign rownumber to each
#subselect for the refGene rows with a name2 that matches with the genes in the
```

```
gene_names.txt file that was converted to a list
#store the select with its subselect as a sql_query and extract selected cols with information from first row of each partitioned group
sql_query="""SELECT name, name2,
      chrom, strand, txStart, txEnd
   FROM (
      SELECT refGene.name, refGene.name2,
      refGene.chrom, refGene.strand, refGene.txStart, refGene.txEnd,
      ROW_NUMBER() OVER (PARTITION BY refGene.name2) AS rn
      FROM refGene
      WHERE refGene.name2 IN ({})
      )
   WHERE rn = 1;""".format(placeholders)
   #.format informs python of placeholders (represented as curly brackets), where the ?'s should be in the sql_query

results=pd.read_sql_query(sql_query, con, params=(*gene_names,)) #pandas expects the placeholders to be ?'s
#provide pandas with sql_query that accepts dynamic variables, db connection, unpacked gene_names
#for the ?'s placed by the format(placeholders), the query runs through the code, replacing the ? with the gene_name

#display dataframe
results
```

Out [4]:

	name	name2	chrom	strand	txStart	txEnd
0	NM_024758	AGMAT	chr1	-	15898193	15911546
1	NM_001375660	ANKRD65	chr1	-	1353799	1356656
2	NM_031921	ATAD3B	chr1	+	1407144	1433228
3	NM_017900	AURKAIP1	chr1	-	1309115	1310587
4	NM_001278054	CASP9	chr1	-	15817895	15850940
5	NM_001791	CDC42	chr1	+	22379201	22427853
6	NM_004442	EPHB2	chr1	+	23037330	23247993
7	NM_004102	FABP3	chr1	-	31838099	31845923
8	NM_012168	FBXO2	chr1	-	11708443	11714486
9	NM_003665	FCN3	chr1	-	27695600	27701315
10	NM_001349683	HSPB7	chr1	-	16340523	16344538
11	NM_005529	HSPG2	chr1	-	22148736	22263803
12	NM_138346	KIAA2013	chr1	-	11979647	11986484
13	NM_001013693	LDLRAD2	chr1	+	22138757	22151716
14	NM_006610	MASP2	chr1	-	11086579	11107296
15	NM_032348	MXRA8	chr1	-	1288070	1293935
16	NM_001278166	NBL1	chr1	+	19974103	19984949
17	NM_007365	PADI2	chr1	-	17393255	17445946
18	NM_001350160	PHACTR4	chr1	+	28696250	28826876
19	NM_000300	PLA2G2A	chr1	-	20301927	20306927
20	NR_146290	PRKCZ-AS1	chr1	-	2113232	2115314
21	NM_052960	RBP7	chr1	+	10057285	10076079
22	NM_178865	SERINC2	chr1	+	31886059	31907525
23	NM_003039	SLC2A5	chr1	-	9095164	9129694
24	NM_001145454	STMN1	chr1	-	26210669	26232910

In [5]:

```
refGenesCoor=pd.DataFrame(results) #save the results as dataframe
refGenesCoor
#load values from refGenesCoor dataframe into sql db; if the data already exists,
replace w new data; w/o index
refGenesCoor.to_sql('refGenesCoor', con, if_exists="replace", index=False)
#write the refGenesCoor dataframe to a csv file with the provided name, providing
the table with header w list of col names
df = refGenesCoor.to_csv('Kalkidan_Tadese_part1.csv', index=False, header=
["name", "name2", "chrom", "strand", "txStart", "txEnd"])
```

In [6]:

```
#to query every table in the db, the table names are stored in a list
# then using a for loop, each table of the list of tables in the dataframe are individually q
#the tissue-specific table with aligned RNA-seq reads replaces the {} since python expect
each placeholder

tbllist =
["burgeRnaSeqGemMapperAlignBrain","burgeRnaSeqGemMapperAlignBreast","burgeRn
    "burgeRnaSeqGemMapperAlignColon","burgeRnaSeqGemMapperAlignHeart",
"burgeRnaSeqGemMapperAlignLiver"]

#initialize empty list to store df resulting from each tissue query
maindf = []

#sql query is stored in sql_query2 variable that contains the query to count reads
#each tissue table contains genomic coordinates that align to different genes in refGene
dataframe
#to compare the individual gene data to the reads of each table, we groupby the gene's r
data to call and compare each count
#join tissue table with new {tissue}_ReadCount with refGenesCoor table
for x in tbllist:
    y = x[25:] #chop off the first part of the tissue name 'burgeRnaSeqGemMapperAlign'
counts as Brain_ReadCount later
    sql_query2="""SELECT refGenesCoor.name, refGenesCoor.name2,
      refGenesCoor.chrom, refGenesCoor.strand, refGenesCoor.txStart, refGenesCoor.
      COUNT(*) AS {}_readCount
      FROM refGenesCoor
      JOIN {} ON
      refGenesCoor.txStart <= {}.chromStart and
      refGenesCoor.txEnd >= {}.chromEnd and
      refGenesCoor.chrom = {}.chrom
      GROUP BY  refGenesCoor.name, refGenesCoor.name2,
            refGenesCoor.chrom, refGenesCoor.strand,
            refGenesCoor.txStart, refGenesCoor.txEnd ;""".format(y,x,x,x,x) #.format in
python of placeholders (represented as curly brackets)
    maindf.append(pd.read_sql_query(sql_query2,con))
#placeholders - for the first curly bracket - y = have it name new "Tissue"_ReadCount, x's
to db table
  #run sql query, append result as df to maindf list


# combines resulting tissue dataframes in maindf list into a single dataframe
ReadCount_df = pd.concat(maindf)
```

In [7]:

```
ReadCount_df
```

Out [7]:

```
         name     name2 chrom strand   txStart     txEnd  Brain_readCount  \
0  NM_001013693   LDLRAD2  chr1      +  22138757  22151716           51.0
1  NM_001145454     STMN1  chr1      -  26210669  26232910         4091.0
2  NM_001278054     CASP9  chr1      -  15817895  15850940           84.0
3  NM_001278166      NBL1  chr1      +  19974103  19984949         1859.0
4  NM_001349683     HSPB7  chr1      -  16340523  16344538           15.0
..          ...       ...   ...    ...       ...       ...            ...
20    NM_032348     MXRA8  chr1      -   1288070   1293935            NaN
21    NM_052960      RBP7  chr1      +  10057285  10076079            NaN
22    NM_138346   KIAA2013  chr1      -  11979647  11986484            NaN
23    NM_178865   SERINC2  chr1      +  31886059  31907525            NaN
24    NR_146290  PRKCZ-AS1  chr1      -   2113232   2115314            NaN

   Breast_readCount  Adipose_readCount  Colon_readCount  Heart_readCount  \
0               NaN                NaN              NaN              NaN
1               NaN                NaN              NaN              NaN
2               NaN                NaN              NaN              NaN
3               NaN                NaN              NaN              NaN
4               NaN                NaN              NaN              NaN
..              ...                ...              ...              ...
20              NaN                NaN              NaN              NaN
21              NaN                NaN              NaN              NaN
22              NaN                NaN              NaN              NaN
23              NaN                NaN              NaN              NaN
24              NaN                NaN              NaN              NaN

   Liver_readCount
0              NaN
1              NaN
2              NaN
3              NaN
4              NaN
..             ...
20            54.0
21            50.0
22           338.0
23          1981.0
24             4.0

[149 rows x 12 columns]
```

In [8]:

```
#pandas fillna to handle null values, replacing the NA for 0
#fillna returns new df; when parameter set to true, fillna replaces NA values in
original dataframe
ReadCount_df.fillna(0, inplace=True)
#display dataframe with replaced NA values
ReadCount_df
```

Out [8]:

```
         name     name2 chrom strand   txStart     txEnd  Brain_readCount  \
0  NM_001013693   LDLRAD2  chr1      +  22138757  22151716           51.0
1  NM_001145454     STMN1  chr1      -  26210669  26232910         4091.0
```

```
2  NM_001278054     CASP9  chr1    - 15817895  15850940         84.0
3  NM_001278166      NBL1  chr1    + 19974103  19984949       1859.0
4  NM_001349683     HSPB7  chr1    - 16340523  16344538         15.0
..          ...       ...  ...  ...       ...       ...           ...
20    NM_032348     MXRA8  chr1    -  1288070   1293935          0.0
21    NM_052960      RBP7  chr1    + 10057285  10076079          0.0
22    NM_138346   KIAA2013  chr1    - 11979647  11986484          0.0
23    NM_178865    SERINC2  chr1    + 31886059  31907525          0.0
24    NR_146290  PRKCZ-AS1  chr1    -  2113232   2115314          0.0

   Breast_readCount  Adipose_readCount  Colon_readCount  Heart_readCount  \
0               0.0                0.0              0.0              0.0
1               0.0                0.0              0.0              0.0
2               0.0                0.0              0.0              0.0
3               0.0                0.0              0.0              0.0
4               0.0                0.0              0.0              0.0
..              ...                ...              ...              ...
20              0.0                0.0              0.0              0.0
21              0.0                0.0              0.0              0.0
22              0.0                0.0              0.0              0.0
23              0.0                0.0              0.0              0.0
24              0.0                0.0              0.0              0.0

   Liver_readCount
0              0.0
1              0.0
2              0.0
3              0.0
4              0.0
..             ...
20            54.0
21            50.0
22           338.0
23          1981.0
24             4.0

[149 rows x 12 columns]
```

In [9]:

```
# drop duplicate rows from the original dataframe
ReadCount_df.drop_duplicates(inplace=True)
#write the dataframe to a named csv file; index not wanted, so index set as false
ReadCount_df.to_csv('Kalkidan_Tadese_part2.csv', index=True)
```

In [10]:

```
#store the ReadCount/result columns in a new list
tissue_columns = [col for col in ReadCount_df.columns if "_readCount" in col]
df_1NF = ReadCount_df.melt(
    id_vars=["name2", "chrom", "strand", "txStart", "txEnd"], #keep gene coors info,
including gene name, name2 etc
    value_vars=tissue_columns, #melt tissue columns; ReadCount is a repeat
group/bin
    var_name="Tissue", #store the melted column in a column called Tissue and
store the expression values in a column called ReadCount
    value_name="ReadCount")
```

```
df_1NF #display melted dataframe
# post-melt modification - remove the _readCount suffix from tissues
df_1NF["Tissue"] = df_1NF["Tissue"].str.replace("_readCount", "")
df_1NF = df_1NF.loc[df_1NF["ReadCount"] != 0] #select rows with non-zero values
in the ReadCount column
df_1NF = df_1NF.sort_values(by="name2",
ascending=True).drop_duplicates().reset_index(drop=True)
#sort the select non-zero readcount rows by their name2 in asc order
(alphabetically)
#reset index without inserting it as a column in a new dataframe
#drop=False would convert row indices into a column

#display sorted/modified dataframe
df_1NF
```

Out [10]:

```
     name2 chrom strand   txStart     txEnd   Tissue  ReadCount
0    AGMAT  chr1      -  15898193  15911546    Heart       1.0
1    AGMAT  chr1      -  15898193  15911546    Liver    1048.0
2    AGMAT  chr1      -  15898193  15911546   Breast      25.0
3    AGMAT  chr1      -  15898193  15911546  Adipose       7.0
4    AGMAT  chr1      -  15898193  15911546    Colon      71.0
..     ...   ...    ...       ...       ...      ...       ...
144  STMN1  chr1      -  26210669  26232910    Heart     136.0
145  STMN1  chr1      -  26210669  26232910    Liver      26.0
146  STMN1  chr1      -  26210669  26232910    Colon     331.0
147  STMN1  chr1      -  26210669  26232910   Breast     123.0
148  STMN1  chr1      -  26210669  26232910  Adipose     180.0

[149 rows x 7 columns]
```

In [11]:

```
df_1NF.to_sql('CleanReadCountTbl', con, if_exists="replace", index=False)
# to_sql() function to write records stored in df_1NF dataframe to a sql database,
provided the connection with con
#if_exists parameter ensures that the table is dropped and created anew if the
table already exists
#Index = False; parameter indicating that we do not want to create a table index
with a new column added for it
```

Out [11]:

```
149
```

In [12]:

```
#saving the modified dataframe to another variable for part 3
df2 = df_1NF

#defining function to calculate mean and standard deviation --> to calculate z-
score
#x.mean and x.std are pandas built-in functions; using paranthesis to order
operations
#z-score = (individual data point - mean)/standard deviation - calculating z-score as
means of normalizing expression values
def mean_time_series(x):
    return (x-x.mean())/x.std()
```

```
# storing the expression values in this "expvalues" col of the dataframe
#grouping by name2 to calculate cumulative statistics of the ReadCount values --
calculate mean, stddev for the read counts of a given gene
df2['expvalues'] = df_1NF.groupby("name2")
["ReadCount"].transform(mean_time_series)
#transform applies row by row - on a #/collection of rows
#df2.drop(columns = "mean")

#reshape dataframe - pivots dataframe from long to wide format
df2_pivoted = df2.pivot(index = ["name2", "chrom", "strand", "txStart", "txEnd"],
values = 'expvalues', columns = 'Tissue').reset_index()
#name2 is how it indexes the new pivoted table (once the tissue column is
unmelted -- one column ==> 6 columns) -- keep this fixed as a reference (index
expects individual column or list of columns)
#values = columns we want to keep the same
```

In [13]:

```
df2_pivoted.dtypes
#display the datatypes of dataframe
```

Out [13]:

```
Tissue
name2      object
chrom      object
strand     object
txStart    int64
txEnd      int64
Adipose    float64
Brain      float64
Breast     float64
Colon      float64
Heart      float64
Liver      float64
dtype: object
```

In [14]:

```
GeneNorm=pd.DataFrame(df2_pivoted) #save the results as dataframe GeneNorm
#write the GeneNorm dataframe to a csv file with my name, do not add an index
df = GeneNorm.to_csv('Kalkidan_Tadese_part3.csv', index=False)
```

In [15]:

```
#import seaborn library for data visualization
import seaborn as sns

#melt -- reshape from wide to long format - individual tissue columns are
converted to variables
df_1NF = df2_pivoted.melt(
   id_vars=["name2", "chrom", "strand", "txStart", "txEnd"],  # Keep only relevant
gene info columns
   value_vars=["Adipose", "Brain", "Breast", "Colon", "Heart", "Liver"],  # Melt only
tissue-related columns
   var_name="Tissue",  # Convert column names into "Tissue" values
   value_name="Expvalues"  # Store z-score
)
```

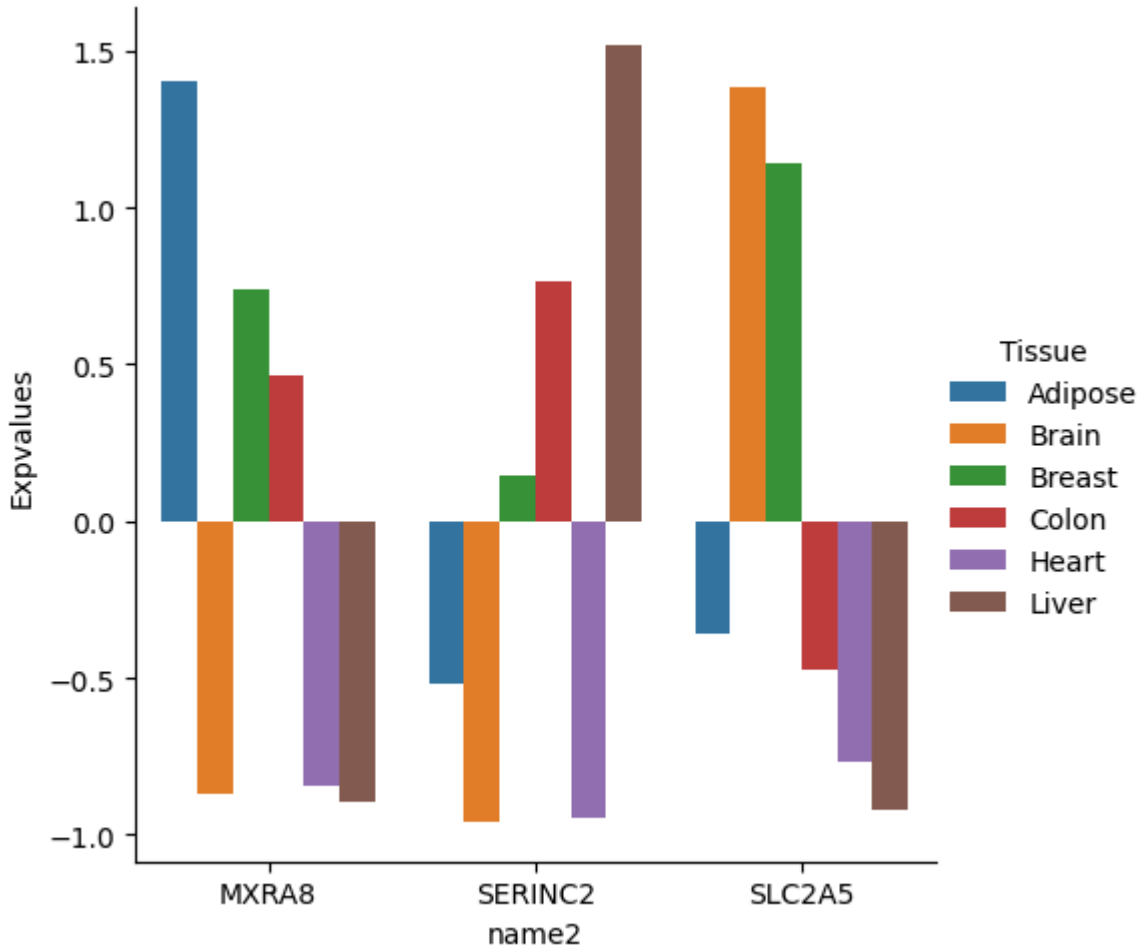
```
#display modified df
df_1NF
```

Out [15]:

```
       name2 chrom strand   txStart     txEnd   Tissue  Expvalues
0      AGMAT  chr1      -  15898193  15911546  Adipose  -0.453977
1     ANKRD65  chr1      -   1353799   1356656  Adipose  -0.246570
2     ATAD3B  chr1      +   1407144   1433228  Adipose   0.190883
3    AURKAIP1  chr1      -   1309115   1310587  Adipose   1.379095
4      CASP9  chr1      -  15817895  15850940  Adipose   0.739048
..       ...  ...    ...       ...       ...     ...       ...
145  PRKCZ-AS1  chr1      -   2113232   2115314    Liver  -0.891534
146      RBP7  chr1      +  10057285  10076079    Liver  -0.543593
147    SERINC2  chr1      +  31886059  31907525    Liver   1.515080
148     SLC2A5  chr1      -   9095164   9129694    Liver  -0.921533
149      STMN1  chr1      -  26210669  26232910    Liver  -0.490289

[150 rows x 7 columns]
```

In [16]:

```
# store in plotgens a list of the genes we want to visualize
plotgens = ["MXRA8", "SERINC2", "SLC2A5"]

#filter dataframe for rows with genes that have name2 that is in the target genes
list
fdf = df_1NF[df_1NF["name2"].isin(plotgens)]
#first [] produces boolean array in dataframe when condition is true/false
#second [] to df filters for the true instances

#create a categorical bar plot comparing normalized expression across tissues
genebarplt = sns.catplot(data=fdf, x="name2", y="Expvalues", hue='Tissue',
kind="bar")
#save barplot as png
genebarplt.savefig("Kalkidan_Tadese_plot.png")
```



In [17]:

```
con.close()
```